

C Thomas, PhD

(they/them)

[Website](#) [in LinkedIn](#) [GitHub](#) [Scholar](#) [email](#) [SF Bay Area](#)

Professional Summary

- **AI R&D Leader bridging frontier science and real-world constraints.** Open to Head/Director of Applied AI Research, or high-impact research leadership focused on frontier model training, evaluation, and interpretability.
- **Expert in bridging research and engineering cultures,** aligning technical and executive teams to achieve high-stakes milestones across academia, startups, and enterprise. Consistently identify gaps and build new teams to solve them.
- **Built and lead an applied AI R&D function** of 25+ contributors across 6 research threads, standardizing core ML training frameworks utilized across hundreds of critical global deployments.
- **Govern massive annual compute allocations and a seven-figure global academic portfolio** to turn early-stage research into standardized, enterprise-wide capabilities across multimodal systems.
- **Cut operational costs by 60% and accelerated model deployment from weeks to hours** by bringing engineering and science into one team, grounding work in factory realities, and driving end-user adoption.

Skills & Tools

R&D Leadership:	Research-to-Production Translation, Cross-Functional Strategy, Team Vision & Engagement, Academic Partnerships, Mentorship
AI/ML Methods:	LLM Pretraining, Computer Vision & VLMs, Domain Adaptation, LoRA Fine-Tuning, Synthetic Data Augmentation, Model Evaluation & Benchmarking
Engineering & Scale:	Systems Engineering, MLOps Infrastructure
Tools:	Python, PyTorch, Hugging Face, LLM Foundry, Detectron2, OpenMMLab

Experience

Apple Cupertino, CA

Applied Research Manager, Machine Learning

2022 – current

- **Built and lead the applied ML research function,** standardizing training frameworks across multimodal initiatives to eliminate isolated silos and enable 20+ MLEs and 5+ cross-functional teams to leverage research breakthroughs across hundreds of global deployments.
- **Scaled organizational research capacity** by unifying ad-hoc university collaborations, guiding 3 internal PIs launching external initiatives, and leading an 80+ person summit to secure broad alignment and research buy-in.
- **Direct a seven-figure annual academic partnerships** and compute allocation across 6 parallel research threads, managing an ecosystem of 14 internal leads and ICs coupled with 4 external faculty and 15+ PhD researchers.
- **Fostered career growth over 5+ years of ML leadership,** leading teams of researchers and engineers, mentoring students and guiding fellow tech leads to clarify intended impact, success metrics, and early progress signals.
- **Closed-loop R&D strategy impacts** include reducing manufacturing-specific training data requirement for synthetic defect generation from 10,000 to just 5 images and cutting overall annotation costs by 60%.
- **Engineered the evaluation pipeline for a novel memory-augmented LLM** (ICLR 2026); authored a patch to LLM Foundry, built the memory-retrieval routing logic, and executed end-to-end evaluations comparing baseline and augmented architectures.

Machine Learning Engineer

2020 – 2022

- **Reduced urgent model deployment time by 80%** by implementing MLOps infrastructure; bridged historical divides between ML, Data Engineering, and Software teams to build a unified data management platform.
- **Trained production computer vision models** for deployment on factory lines, building monitoring systems to detect model drift and core vulnerabilities prior to customer impact.
- **Established engineering standards** and mentorship to junior ICs, interns and student collaborators in collaborative software development and rigorous experimental data interpretation.

Tua Fertility Oakland, CA

Founder & CEO

2019 - 2020

- **Designed, patented and prototyped a medical device,** executing end-to-end mechanical design (CAD/SolidWorks) and rapid iterations to proof of concept as a resident at the Autodesk Technology Center
- **Drove technical vision and business strategy,** pitching to investors and translating complex engineering roadmaps into compelling narratives.

Rigetti Computing Berkeley, CA

Senior Research Engineer & Tech Lead

2017 - 2019

- **Navigated early-stage startup ambiguity to drive critical technical breakthroughs** by aligning teams and sub-systems to stabilize and optimize complex, non-linear computing architectures.
- **Built robust R&D software** to calibrate and run quantum gate operations, analyzing large datasets and distilling results into shared tooling for research teams.
- **Influenced executive technical roadmaps** by ruthlessly prioritizing actionable R&D threads, translating complex engineering progress into actionable milestones that shifted company timelines and targets.

University of California, Berkeley Berkeley, CA

Doctoral Research Fellow

2010 - 2017

- **Awarded the highly competitive DOE SCGF (accepted) and NSF (declined) Fellowships.**
- **Designed and built high-precision optics, hardware and electronic systems** to trap individual atoms in lattices of laser light. Drove quantum phase transition and explained dynamics with first-principles mean-field model ([Phys. Rev. Lett., Editor's Suggestion](#)).
- **Resolved a contested phenomenon in quantum literature** by developing a predictive white-box model that provided a clear explanation for unexpected system behavior ([Phys. Rev. A](#)).

Selected Publications

Pretraining with hierarchical memories: separating long-tail and common knowledge.

H. Pouransari, D. Grangier, [C. Thomas](#), M. Kirchhof, O. Tuzel. *ICLR 2026*. [[arXiv](#)]

Rethinking Semi-Supervised Domain Adaptation for Semantic Segmentation in the Era of Foundation Models. J. Kurien*, B. Balaji*, K.H. Lai, P. G. Vela, [C. Thomas](#), A. Wong, S. Rambhatla. *CVPR 2025 (FGVC Workshop)*.

LangDA: Language-guided Domain Adaptive Semantic Segmentation. C. Liu, S. Hossain, [C. Thomas](#), K.H. Lai, R. Vemulapalli, S. Rambhatla, A. Wong. *NeurIPS 2024 (AFM Workshop)*. [[NeurIPS](#)] [[arxiv](#)]

Synth4Seg: Learning Defect Data Synthesis for Defect Segmentation using Bi-level Optimization. S. Mou, R. Vemulapalli, S. Li, Y. Liu, [C. Thomas](#), M. Cao, H. Bai, O. Tuzel, P. Huang, J. Shan, J. Shi. *IEEE T-ASE 2024*. [[arXiv](#)]

Service & Leadership

Academic Conference Leadership

Co-organizer, 4th Workshop on Vision-based Industrial Inspection , CVPR	2026
Invited Speaker, "Applied Research in Industrial AI", 3rd Workshop , ICCV	2025
Co-organizer & Panelist, Data-efficient defect segmentation, 2nd Workshop , ECCV	2024

Career Mentorship

Committee Member, Employee Career Development & Rotations, Apple Operations	2023 – Current
Career Mentor, 1:1 guidance for recent graduates entering the workforce	2023 – Current

Community

Impact Bay Area is a nonprofit in the SF Bay Area that trains women and girls to set and hold boundaries with verbal and physical self-defense skills.

Board Secretary	2023 – 2025
Assistant Instructor	2017 – 2025

Education

PhD	University of California at Berkeley, Atomic and Optical Physics Quantum simulation of triangular, honeycomb and kagome crystal structures using ultracold atoms in lattices of laser light	2017
MA	University of California at Berkeley, Physics	2013
BA	Boston University, Physics and Math	2010